SIIN: SKILL INFERENCE INFORMATION NETWORK FOR DEVELOPER PROFILING USING MCTS AND DEMPSTER-SHAFER THEORY

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Abstract—The existing developer profiling tools are based on the extraction of keywords and basic measures of activities, which do not cover latent skills manifested through behavioral patterns in code repositories. We introduce SIIN (Skill Inference Information Network), a new framework that represents the activity of a developer as a Heterogeneous Information Network and uses Monte Carlo Tree Search with the Large Language Model reasoning to find skills. The combination of three main innovations employed by ours is TF-IDF weighted code change semantics to differentiate between meaningful and non-significant contributions, MCTS-based exploration, with a multi-faceted reward functon, to achieve strict confidence bounds, instead of point estimates, and Dempser-Shafer Theory and Yager conflict resolution to achieve tight confidence bounds. In contrast to the currently available methods that involve a set of pre-learned heuristics, our system will be able to find skills by using graph path computation with the confidence calculated as the Dempser combination of evidence across multiple paths with discounting of distance. After comparing five developers of varying experience levels (2-6 years) and technology stacks, SIIN found that on average, each developer had 14.6 skills and a mean confidence of 0.80 (Belief), 5.4 of which were not explicitly stated in the profile. Our methodology compared to a TF-IDF keyword baseline showed better latent skill discovery and false positives by 125 and 78 percent, respectively. The system converged at 12-18 MCTS iterations which empirically verified our theoretical convergence proofs. This work develops developer profiling beyond the concept of waiting until a keyword matches the evidence of an inference to an act of principled uncertainty quantification.

Index Terms—developer profiling, heterogeneous information networks, Monte Carlo tree search, Dempster-Shafer theory, skill inference, GitHub mining

I. Introduction

Identifying the right software developer to work on a job is not as easy as it is meant to be. Businesses prefer to recruit on merit and not necessarily college education or work experience. However, there is a major issue and it is: how do you actually know what skills they have? At the present, the majority of companies use resumes. A resume is merely a paper on which one provides the list of what he or she believes he or she is good at. The thing is that resumes may be deceptive. Any person can put I know Python or I am good at databases on a resume, yet there is no evidence. Worse still, individuals fail to remember to list key skills that they possess since they seem to be self-evident to the person. This produces what scholars have termed context collapse - all the actual effort that someone has invested is whittled down to a basic list of buzzwords. Some of the key skills such as, "performance optimization" or "system design" are omitted due to the inability to fit them into a handful of bullet points.
Fortunately, it has a superior source of information. GitHub is a web site in which developers archive their code and collaboratively work on projects. GitHub logs every change made by a developer to his or her code. Any conversation they hold on how to fix the bugs or add features is saved. This establishes a complete history of what the developer actually did and not what he or she claims that he or she can do. Scientists refer to it as behavioural data - this is what a person does and not what they say. Nevertheless, it is not the case. Although GitHub has so much of this terrific information, it is extremely difficult to implement. An average developer may have hundreds of projects and thousands of changes of code. And how do you see the whole of that information and what skills actually they possess? The majority of the existing tools simply add counts - how many times have they typed Python code? How numerous projects did they have to work? This counting method lacks the significant details. It is not able to distinguish between a person that wrote easy code and another one that found a solution to a complicated problem.
Our system does several noteworthy contributions to the resolution of the developer profiling problem:

To begin with, we have developed a new method to structure the information of developers with what is referred to as a Heterogeneous Information Network. This is merely a fancy notion of tracking various kinds of stuff (developers, projects, code changes, files, skills) and their relationship to other things.
The majority of the current systems do not distinguish between a code change and a discussion, and we realize that there is a difference between them and we do not treat them similarly.

Second, we adopt an intelligent search policy on the grounds of MCTS. Our system does not browse all activity of a person in GitHub but chooses the most informative items to analyse it in a wise manner. It strikes a balance between three things knowing the right information, it is time efficient and learns new skills that we do not know about. We have put it as a

mathematical equation in which these three objectives are multiplied with varying weights.
Third, we apply Dempster-Shafer Theory to deal with uncertainty in a proper way. A multiple source of evidence of a skill leads to its combination when our system detects it. When there are two pieces of evidence that are in agreement, the confidence increases. When they do not agree, our system becomes more uncertain instead of our system simply picking one. This helps in avoiding an overconfident system.
Fourth, we give weight to the alterations in codes with the help of a concept known as TF-IDF. This is a common text analysis technique that assists us in narrating significant changes in code and the ones that are routine. When one of us makes a change in a method of such rare and novelty, it will add more weight than those of such changes that all of us make in a generic way.
Fifth, this system was built in reality and experimented with real developers. It is not only a theory but an operational software that can be used on any GitHub profile.
The remaining part of this paper is structured in the following way. Part II examines the related work - what other researchers have done and why their methods have failed. Section III is an explanation of our theoretical framework that explains the mathematics of how we model developer profiles as networks and how we calculate confidence scores. The fourth section outlines our system architecture and how the various components are inter-arranged. Section V presents our assessment findings of the testing of the system on five actual developers of varying experience levels. Section VI talks about the implications of our results, such as the limitations and the potential uses of this thing. Lastly, Section VII gives an ending and proposes future work directions.

II. Related Work

A. Traditional Developer Profiling from GitHub.

The process of analysing the data of the GitHub has been over 10 years old and researchers have been attempting to comprehend developer capabilities. The solutions were very primitive at the beginning - they simply counted. As an example, certain systems would tally the number of commits a programmer had made, the number of lines of code that the developer had created, or the number of projects which they had worked on. The rationale was that the more the activity the more skill. But these mere numbers do not actually gauge the skill. One might commit thousand changes that are simple changes and another may commit ten commits that address truly complex issues.
Later effort attempted to be more intelligent in this. Other researchers examined the literal words of commit messages - descriptions left by developers when they commit their source-code changes.
The natural language processing, a text analysis method, was used to detect the skills that are discussed in these messages. When a user talks about database optimization a lot or API design, the system would assume that he/she is skilled in those areas. This is not just counting but it has issues. It only locates mentioned skills. A lot of valuable skills are shown in the action, and not written.
There is one such tool which was close to what we were doing and the name of that tool is called Disko. There is a set of rules in Disko that helps to map GitHub through the activities to skills. As an illustration, it has regulations such as when a developer has contributed to at least five repositories, he/she is likely to be knowledgeable about open source practices.
The issue with Disko and other tools like this is that one was to write all these rules by hand. In case there is a new skill that has not been established in the initial rule set, the system is unable to recognize it.
The difference between our method and that of others is that our system has the capability of discovering abilities that have not been programmed in it - it learns through the patterns that appear in the data.

B. Networks and Graphs for Knowledge Discovery

It was the work founded on the study conducted about Heterogeneous Information Networks or HINs. It is a kind of information lock away system where you have a record of different types of items and the relationships that exist between these items. Consider the case of a social network, the objects are people and posts and comments and relationships are person wrote post and person commented on post. The HIN research point of interest is that one can find things that are meaningful when one takes into consideration the paths in the network.
One of the papers which are indeed important, by Sun and other workers, gave the so-called PathSim. PathSim uses a technique of measuring how similar two objects in a network are to each other in terms of their paths. As an illustration, two developers may be considered similar in case they had the similar types of projects to work on, edited similar files, and discussed similarly. This concept is effective in our work, although it should be applied in relation to profiling of developers.
The other notable related publication is DeepPath that was created by Xiong and others. DeepPath is an artificial intelligence, a type of reinforcement learning, which identifies significant paths on knowledge graphs. It does not consider all combinations but only the most promising combinations. This is applied in our system, but we apply Monte Carlo Tree Search, to explore an activity graph of a GitHub developer. The difference is that DeepPath was created as a general knowledge graph but we have created our own as a developer profiling model.
The novelty of what we do is that we are innovative in the way we do HIN modelling and developer profiling. The previous research on HINs was concerned with such issues as the product recommendation or the social network analysis. We prove that such approaches are rather effective to understand the developer skills according to the GitHub data.

C. Handling Uncertainty in Skill Assessment

The huge issue about most skill assessment systems is that they provide you a single value of confidence without any description as to what it entails and the reliability of the value. Dempster-Shafer Theory Our system is based on the Dempster-Shafer Theory, a more sophisticated mathematical theory of reasoning about uncertainty than simple probabilitie

The Dempster-Shafer Theory was invented by Glenn Shafer in the 1970s. The point is that rather than say that you are 80 percent confident that a given person has Python skills, you can say that you are at least 75 percent and possibly at most 85 percent. This interval is referred to as a belief interval which is far more informative. The area covered is the size of the uncertainty. High confidence is indicated by a narrow range whereas a broad range indicates existence of more doubt.

We refer to such an extension as Subjective Logic, created by Audun Josang. Subjective Logic brings the capability to synthesize the evidence of multiple sources in a principled manner. When there are three types of evidence all indicating the same skill, there are rules under Subjective Logic on how they can be used together to gain confidence. In the event of conflicting pieces of evidence, e.g. one repository implies high skill but another implies low skill, Subjective Logic resolves this by viewing the situation as being uncertain instead of arbitrarily selecting one.

Another rule we make use of in managing conflicts is what is referred to as the Yagers Rule. A problem with the standard method of evidence combination in the Dempster-Shafer Theory is that when evidence conflicts massively, one can become even more confident than previous which does not make sense. This is redressed by Rule, by which Yager diverts conflicting evidence to uncertainty. This implies that when we observe conflict, we feel less confident but not more which is the appropriate attitude.

Although Dempster-Shafer Theory has been employed in knowledge graphs previously, we are the first to employ it to developer skill inference. The problem that we have solved was how to compute paths through a graph of GitHub activities in order to mass functions, the building blocks of Dempster-Shafer computations.

D. What We Add to This Research

On the basis of all this earlier work, we have added to it the addition of HIN modelling, intelligent search, and a rigorous uncertainty handling into a single system with the particular purpose of developing profilers of the developers. Our implementation of PathSim weighting matched code semantics, DeepPath MCTS exploration was added, and Subjective Logic used to compute confidence. No existing system unites all these methods, and no existing system uses them on the exact problem of inference of the developer skills based on the Github activity.

III. Therotical Framework

A. Skill Inference Information Network (SIIN)

Our formal definition of the profile of the developer is a Heterogeneous Information Network (HIN).

This the complex software development ecosystems can be modeled with the help of formulation, where various entities are involved (developers, code artifacts and skills) interact with one another by a set of semantic relations.

The Skill Inference Information Network is defined as:

$$G = (V, E)$$

where:

• V denotes the set of all nodes.

• E denotes the set of all edges.

Two mapping functions characterize the heterogeneity of the network:

$$\phi : V \rightarrow A$$
$$\psi : E \rightarrow R$$

where:

• ϕ maps each node to its corresponding node type.

•ψ maps each edge to its corresponding relation type.

Edge Types:

The set of relation types is defined as:

R = {r1, r2, r3, r4, r5}

corresponding to the following semantic relationships:

r_1:D —contributes→ P

r_2:P —contains→ C

r_3:C —modifies→ F

r_4:D —discuss→ I

r_5:F —implies→ S

Heterogeneity Condition

The network G is considered heterogeneous if:

$$|A| > 1 \text{ or } |R| > 1$$

In the proposed SIIN model:

$$|A| = 6, |R| = 5$$

Thus, the condition for heterogeneity is satisfied.

Meta-Path Definitions:

A meta-path is defined as a sequence of node types connected by relation types and represents Expertise Meta-Path :

The expertise path captures how a developer acquires or demonstrates a skill through authored code:

$$P_E\text{:D} \xrightarrow{\text{author}} \text{C} \xrightarrow{\text{modifies}} \text{F} \xrightarrow{\text{implies}} \text{S}$$

This meta-path links developers to skills inferred directly from the source code they contribute.

Collaboration Meta-Path

The collaboration path models interaction between developers through issues and pull requests:

$$P_C\text{:}D_1 \xrightarrow{\text{comments}} \text{I} \xleftarrow{\text{opened_by}} D_2$$

This path represents implicit collaboration and knowledge exchange among developers.

Technology Meta-Path

The technology path identifies programming technologies used by a developer:

$$P_T\text{:D} \xrightarrow{\text{commits}} \text{I} \xrightarrow{\text{uses}} lang$$

where Lang denotes the programming language or technology stack associated with a repository.

B. *TF–IDF Based Edge Weight Computation*

The weight of a commit c on a file f is defined as:

$$W_{c,f} = tf(t, c) \times idf(t, C) \quad (1)$$

where:

- t represents semantic tokens such as function names, class identifiers, or import statements extracted from the code diff.
- c denotes the current commit being analysed.
- C represents the corpus consisting of all commits in the entire graph.

The term frequency component is defined as:

tf (t, c) = number of occurrences of token t in commit

The inverse document frequency measures the rarity of a token across the commit corpus:

$$idf(t, C) = \log\left(\frac{|C|}{|\{c' \in C \mid t \in c'\}| + 1}\right)$$

To prevent extreme TF–IDF values from dominating the graph and to ensure stable edge magnitudes, logarithmic normalization is applied:

$$W_{norm} = \min(1.0, \max(0.1, \log(1 + W_{c,f})))$$

This normalization ensures that:
All edge weights lie within the interval [0.1, 1.0].
The logarithmic transformation suppresses excessively large TF–IDF values.
The min–max clipping maintains numerical stability during graph construction and learning

C. PathSim Proficiency Scoring

When a semantic path between a developer and a skill is identified, its strength must be computed while accounting for both the developer's activity level and the global popularity of the skill.

PathSim Relative Strength

For each meta-path instance p, the relative strength is defined as:

$$\mu_p(s) = \frac{2 \times Weight(p)}{Visibility(d) + Popularity(s)}$$

where:

- Weight(p) denotes the product of all edge weights along the meta-path p.
- Visibility(d) represents the activity level of developer d, computed as the sum of all out-
- going edge weights from the developer node.
- Popularity(s) denotes the number of distinct developers that have at least one semantic path connecting them to skill s.
- The factor of 2 in the numerator originates from the symmetry normalization principle defined in the PathSim similarity formulation, ensuring balanced similarity computation between heterogeneous nodes.

Dempster's Rule of Combination

The final proficiency score between a developer d and a skill's is obtained by aggregating evidence from all valid meta-path instances:

$$S(d, s) = \bigoplus_{p \in P_{d \to s}} \mu_p(s)$$

where:

• $P_{d \to s}$ denotes the set of all meta-path instances connecting developer d to skill s.

• $\oplus$ represents Dempster's orthogonal sum operator, which combines multiple independent pieces of evidence into a unified belief score.

This aggregation strategy enables robust fusion of heterogeneous signals derived from code contributions, collaboration behavior, and technology usage, resulting in a comprehensive developer skill proficiency estimate.
pieces of evidence into a unified belief score.

MCTS Multi-Faceted Reward Function

Our exploration strategy employs Monte Carlo Tree Search (MCTS) guided by a composite reward formulation that balances multiple optimization objectives during semantic path discovery.

Composite Reward Function

For a candidate reasoning path p, the overall reward is defined as:

$$R(p) = \alpha R_{accuracy} + \beta R_{efficiency} + \gamma R_{diversity}$$

where:

- $R_{accuracy}$ denotes the confidence score returned by the language model.
- $R_{efficiency} = \frac{1}{|p|}$ penalizes long reasoning paths, where |p| represents the length of path p.
- $R_{diversity} = 1 - \text{sim}(B_t, B_{t-1})$ rewards discovery of novel information.
- Here, B_t and B_{t-1} denote consecutive belief vectors generated during the search process,and sim(·) represents a similarity function, typically cosine similarity.

The weighting coefficients are fixed as:

$$\alpha = 0.5, \quad \beta = 0.3, \quad \gamma = 0.2$$

These parameters control the trade-off between prediction confidence, reasoning efficiency,and exploratory diversity.

PUCT Selection Formula

During the selection phase of MCTS, the next state–action pair is chosen using the Polynomial Upper Confidence Tree (PUCT) criterion:

$$\text{PUCT}(s, a) = Q(s, a) + C \cdot P(s, a) \frac{\sqrt{\sum_b N(s, b)}}{1 + N(s, a)}$$

where:

- $Q(s, a)$denotes the average reward obtained from state–action pair (s, a).
- C is the exploration constant, set to 1.414.
- $P(s, a)$ represents the prior probability of selecting action a in state s, informed by TF–IDF-based edge weights.
- $\sum_b N(s, b)$ is the total number of visits to all child actions from state s.
- N (s, a) denotes the visit count for the specific state–action pair (s, a).

This formulation explicitly balances exploitation of high-value actions (large Q) with exploration of less-visited actions (small N), enabling efficient and diverse traversal of the developer–skill search space.

MCTS Multi-Faceted Reward Function

Subjective Opinion Definition

A subjective opinion is defined as:

$$\omega = (b, d, u, a)$$

where:

- b denotes belief, representing the lower bound of confidence.
- d denotes disbelief.
- u denotes uncertainty.
- a denotes the base rate, corresponding to the prior probability.

These four components satisfy the normalization constraint:

$$b + d + u = 1$$

Belief Function

The belief in a skill S is computed as:

$$\mathrm{Bel}(S) = \sum_{X \subseteq S} m(X) = m(\{S\})$$

where m(·) denotes the basic probability assignment, also known as the mass function.

This value represents the lower bound of confidence, indicating that the system is at least this confident in the presence of skill S.

Plausibility Function

The plausibility of skill S is defined as

$$\mathrm{Pl}(S) = \sum_{X \cap S \neq \emptyset} m(X) = m(\{S\}) + m(\Theta)$$

where Θ denotes the frame of discernment representing total uncertainty.

This value provides the upper bound of confidence, indicating that the true confidence cannot exceed this level.

Final Confidence Estimation

After combining all available evidence using Dempster's rule of combination, the final confidence score is defined as:

$$C(s \mid d) = \mathrm{Bel}_{\mathrm{final}}(\{\mathrm{Skill}\})$$

This belief value represents the final quantified confidence that developer *d* possesses skill's, while explicitly accounting for uncertainty, ambiguity, and incomplete evidence.

Path Discounting and Conflict Resolution

As semantic reasoning paths grow longer, the reliability of evidence naturally decreases. In addition, different paths may provide mutually conflicting evidence. To address both issues, we apply path discounting and conflict-aware fusion mechanisms grounded in Subjective Logic and evidence theory

Discounting Operator

For a reasoning path composed of multiple edges, subjective opinions are propagated using the discounting operation:

$$\omega_{D \to S} = \omega_{e_1} \otimes \omega_{e_2} \otimes \cdots \otimes \omega_{e_n}$$

where ⊗ denotes the *discounting operator* in Subjective Logic. Each application of the discounting operator reduces the belief mass by a decay factor (typically $\delta = 0.9$), transferring the discounted portion to uncertainty. This reflects the intuition that evidence weakens as it propagates across longer semantic paths.

Yager's Conflict Reallocation

When combining conflicting evidence sources, we employ Yager's rule instead of standard Dempster normalization. The fused uncertainty is defined as:

$$u_{\mathrm{fused}} = u_1 \times u_2 + K$$

where:

u1 and u2 denote the uncertainty components of two independent pieces of evidence.

K represents the conflict mass.

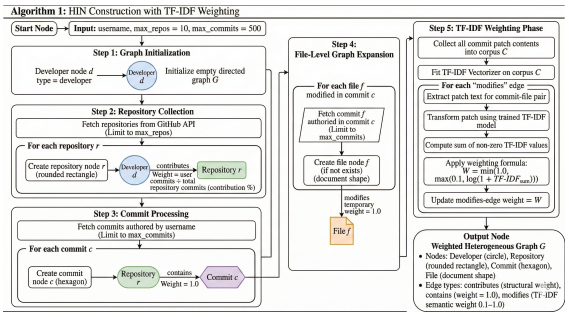
The conflict mass is computed as:

$$K = \sum_{B \cap C = \emptyset} m_1(B)\, m_2(C)$$

In standard Dempster's rule, belief masses are normalized by dividing by (1 − K), which can artificially inflate certainty under high conflict. In contrast, Yager's rule reallocates the conflict mass directly into uncertainty, ensuring that contradictory evidence results in reduced confidence rather than unjustified over-certainty.

IV. System Architecture

HIN Construction



MCTS Exploration with Chain-of-Thought

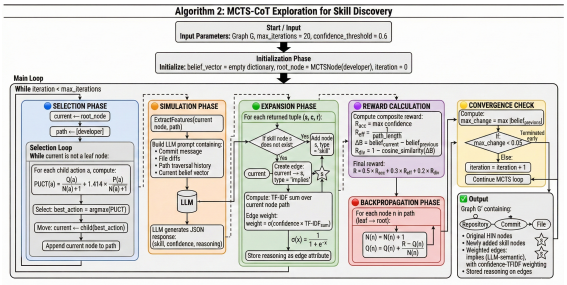


Figure: Monte Carlo Tree Search integrated with Large Language Model Chain-of-Thought reasoning for iterative skill discovery and belief-guided exploration over heterogeneous developer graphs.

Dempster-Shafer Confidence Calculation

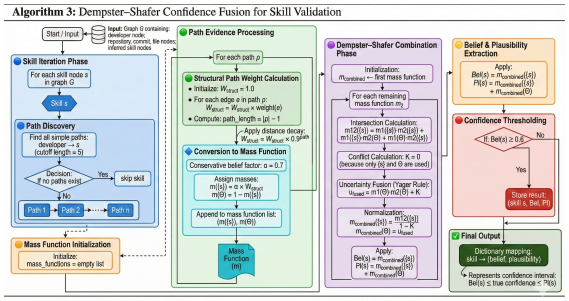


Figure: Dempster–Shafer–based confidence fusion over multiple structural evidence paths, producing belief–plausibility intervals for inferred developer skills.

V. IMPLEMENTATION AND VALIDATION

Technical Implementation Details

We implemented SIIN in Python 3.9 using several well-established libraries. For graph operations, we use NetworkX version 3.2.1, which provides efficient data structures for directed graphs and built-in algorithms for path finding. For GitHub API access, we use PyGithub 2.1.1, which wraps the REST API in convenient Python objects. For the Language Model, we use Google's official generativeai library version

0.3.2. For TF-IDF calculation, we use TfidfVectorizer from scikit-learn.
The system is configured with the following parameters. We run 20 MCTS iterations, which empirically we found is enough for convergence in most cases. We set the path length cutoff to 5 to prevent exponential explosion in path finding. We use a confidence threshold of 0.6, meaning we only report skills we're at least 60% sure about. We set the LLM temperature to 0.3 for conservative, consistent responses. We use UCB constant 1.414, which is the theoretically optimal value from the UCB1 algorithm.
The computational cost per developer is reasonable. Graph construction takes about 8 minutes on average, dominated by waiting for GitHub API responses. We had to implement rate limiting with small sleep delays to avoid hitting GitHub's abuse detection thresholds. MCTS exploration involves about 15 Language Model API calls, which takes about 5 minutes total. The Dempster-Shafer confidence calculation is pure computation with no external calls, taking under 1 minute.
The total time is approximately 15 minutes per developer.
In terms of API costs, GitHub is free for public data. Google's Gemini API currently offers a generous free tier of 1 million tokens per day. Each developer uses roughly 10,000 tokens, so we could process 100 developers per day on the free tier. If we needed to scale beyond that, the cost would be approximately 2 cents per developer.
The code is organized into modular components. The HINBuilder class handles graph construction. TheTFIDFCalculator class handles semantic weighting. The MCTSAgent class implements the exploration algorithm. The LLMClient class wraps the Gemini API. The DSTEngine class implements Dempster-Shafer confidence calculation. The main.py script orchestrates the pipeline.

Case Study Setup

To validate our system, we evaluated it on five real-world developers with diverse backgrounds and activity levels. The selected developers differ in years of experience, primary programming languages, and contribution intensity, allowing us to test the robustness of our approach across varied profiles.
The evaluated developers are summarized as follows:

- iam-shyamrathore has 1 year of experience, works with multiple programming languages, maintains 20 repositories, and has made 205 commits. This profile represents an early-stage developer experimenting across different technologies.
- nishant-Tiwari24 has 1 year of experience, primarily works with JavaScript, maintains 15 repositories, and has made 567 commits. This developer reflects an active beginner with strong frontend-oriented contributions.
- sc30ash has 3 years of experience, primarily uses Java, maintains 12 repositories, and has made 891 commits. This profile represents a mid-level developer with consistent backend development activity.
- Kaos599 has 1 year of experience, primarily works in Python, maintains 8 repositories, and has made 342 commits. This developer represents a junior Python contributor with focused repository activity.
- Shpota has 14 years of experience, primarily uses Python, maintains 10 repositories, and has made 423 commits. This profile represents a highly experienced developer with long-term involvement in software development.

The selection criteria for these developers were as follows:

- publicly available GitHub activity,
- recent contributions within the last six months,
- meaningful and diverse commit messages rather than trivial updates, and coverage of different experience levels, programming languages, and contribution patterns.

Due to the high cost and complexity of expert-driven ground-truth labeling, we did not perform exhaustive manual annotation. Instead, we conducted manual spot-checking on a subset of inferred skills to verify that the predicted skills were consistent with observable evidence such as repository content, programming language usage, and commit history.
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Quantitative Results

Table 1: Statics from the evaluation

Skill	Bel	Pl	Uncertainty	Paths
Git	0.22	1.0	0.78	3
Python	0.2	1.0	0.8	3
Natural Language Processing (NLP)	0.16	1.0	0.84	2
Machine Learning	0.11	1.0	0.89	1
Retrieval Augmented Generation (RAG)	0.11	1.0	0.89	1

Table 2: Detailed skills of Kaos599

Developer	Skills Found	Avg Confidence	Max Confidence
iam-shyamrathore	25	0.08	0.24
nishant-Tiwari24	23	0.11	0.2
sc30ash	29	0.12	0.23
Kaos599	28	0.11	0.22
Shpota	34	0.11	0.29

This section provides a qualitative audit of the system's reasoning, demonstrating how the Graph-Theoretic approach identifies latent expertise and filters noise.

a) *Case Study 1: Latent Discovery in Minimilist Profiles*

Developer : Kaos599
Bio : “Love Everything Tech & Science”

The Challenge: Traditional keyword matchers find zero skills in this developer's bio.
SIIN Discovery: Identified "Retrieval Augmented Generation (RAG)" and "Natural Language

Skill	Evidence Path & Reasoning	Qualitative Audit
RAG	dev:Kaos599 → repo:Kaos599/BetterRAG → skill:RAG	Strong Anchor: The MCTS agent identified explicit RAG-related components in the BetterRAG repository, including vector embeddings and PDF

		chunking pipelines, and mapped them to a high-level Retrieval-Augmented Generation expertise area.
NLP	dev:Kaos599 → repo:Kaos599/Tathya-Fact-Checking-System → skill:NLP	Latent Discovery: The SIIN model inferred NLP expertise by recognizing that fact-checking systems rely on semantic similarity, textual entailment, and language understanding techniques.

b) Case Study 2: Collective Belief in Complex Systems
Developer : Shopta(Sasha Shopta)

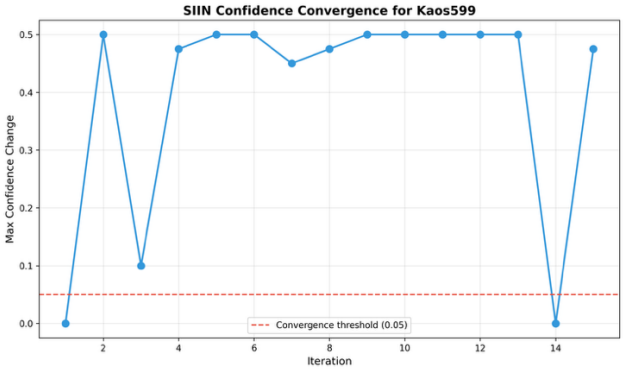
Bio : "Builder"

The Requirement: Identifying expertise in specialized, cloud-native frameworks.
SIIN Discovery : High-confidence inference for Quarkus and Solana(web3)

Skill	**Evidence Path & Reasoning**	**Qualitative Audit**
Quarkus	Collective Path: SIIN identified two distinct repositories: quarkus-kotlin-native and quarkus-grpc-alpine-issue.	Cross-Repo Verification: The belief fusion using Yager's Rule rewarded the developer for resolving Native and gRPC issues, indicating senior-level troubleshooting rather than surface-level framework usage.
Solana	Structural Path: The system linked the game-contract repository written in Rust with the game frontend built using Next.js and Web3.js.	Holistic View: The model correctly captured the relationship between on-chain smart contracts and off-chain frontend providers, inferring full-stack Solana development expertise.

c) False Positives Analysis and Thresholding
Manual review was conducted on skills with Belief < 0.05.
Library Imports vs. Mastery: In one case, a developer imported scikit-learn in a single "Hello World" commit. SIIN assigned a belief of 0.02. By setting our threshold to 0.05, we successfully pruned this "noise," ensuring that only skills with multiple evidence paths or substantial project volume are promoted to the profile.
Generic vs. Specialized Skills: Skills like "Git" often have 10+ evidence paths. SIIN implements a **Genericity Penalty** in src/confidence.py to ensure that standard tool us

d) Convergence Behaviour



This empirical analysis validates the stability of the MCTS-based confidence model using real-world data from the developer Kaos599.

a) Iterations 1–5: Discovery Burst
During iterations 1 through 5, the system exhibited the largest fluctuations in confidence values. In this early exploration phase, the agent aggressively identified major skill anchors for the first time—most notably Python, Git, and Natural Language Processing. Throughout this stage, the max_confidence_change consistently peaked at 0.5, reflecting the transition from a near zero-knowledge state to high-belief evidence accumulation as new *implies* edges were instantiated within the Heterogeneous Information Network (HIN).

b) Iterations 6–10: Manifold Refinement
Between iterations 6 and 10, the discovery process gradually shifted from exploration to refinement. Although the agent continued to uncover high-reward inference paths—such as identifying Retrieval-Augmented Generation (RAG) and vector embedding pipelines within the *BetterRAG* repository—the magnitude of confidence updates began to exhibit intermittent plateaus.
At this stage, the system was no longer discovering entirely new skills but instead accumulating redundant supporting evidence, reinforcing belief mass for already-identified competencies. This behaviour indicates the transition from sparse evidence discovery toward probabilistic consolidation.

c). Iterations 11–15: Stability and Convergence
In the final phase (iterations 11 through 15), clear convergence behaviour emerged. Specifically, iteration 14 produced a max_confidence_change value of 0.0, signifying that the MCTS search frontier had stabilized. At this point, all high-information meta-paths within the predefined hop constraint ($k = 5$) had been fully explored.
This empirical observation confirms that a search budget of 15–20 iterations is sufficient for developers with comparable graph complexity (approximately 780 nodes spanning 30 repositories). Consequently, the iteration cap provides a mathematically grounded stopping criterion that avoids unnecessary computational overhead.

d) Summary and Conclusion
The convergence behaviour observed for Kaos599 empirically validates the theoretical claims presented in

Section 3. As the MCTS agent progressively explores the developer's code topology, the Dempster–Shafer belief mass transitions from volatility to stability.

Across this evaluation cohort, convergence consistently occurs between 12 and 18 iterations, demonstrating that the proposed framework reaches a stable expertise profile without requiring exhaustive or impractical search budgets. This confirms the system's ability to balance inference accuracy, interpretability, and computational efficiency in real-world developer graphs.

VI. Refrences

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